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The impact of political ties on firm innovativeness: Testing a mediation and moderation model

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ABSTRACT

This study utilizes network exploitation and exploration theory to distinguish the exploitative and exploratory mechanisms (government support and boundary-spanning learning, respectively) that convert political ties into firm innovativeness. It also examines the moderating roles market uncertainty plays in how political ties influence firm innovativeness. An empirical analysis of 262 manufacturers in China indicates that both government support and boundary-spanning learning mediate the link between political ties and firm innovativeness. Market uncertainty negatively moderates the impact of political ties on boundary-spanning learning but does not significantly moderate their impact on government support. Moreover, market uncertainty moderates the impact of government support on firm innovativeness negatively but moderates the impacts of boundary-spanning learning on firm innovativeness positively. This study theorizes two mediating factors for how firms exploit and explore their political ties to increase their innovativeness and enriches the knowledge of how market uncertainty affects the relationship between political ties and innovation.

1. Introduction

As a category of social network ties, political ties play a vital role in influencing firms' activities and performance (Peng & Luo, 2000; Zhang, O'Kane, & Chen, 2020). Scholars suggest that political ties, which are defined as top managers' personal connections with government officials, act as strategic options for firms to obtain more resources or opportunities and enhance their regulatory legitimacy in a transition economy (Sheng, Zhou, & Li, 2011; Zhang, Qi, Wang, Zhao, & Pawar, 2019). Despite their declining influence amid the institutional changes in a transition economy (Hitt & Xu, 2016), political ties are still useful for firm innovation, where external resources and knowledge are particularly important (Gao, Shu, Jiang, Gao, & Page, 2017). In China's transition economy, for example, the central government has emphasized developing innovation as a means to overcome a middle-income trap. Chinese government officials maintain close interactions with industries to ensure that China's innovation policies are promoted (Tjosvold, Peng, Chen, & Su, 2008). However, firm innovation activities are accompanied by higher risks when legal and market institutions are weak. Here, political ties are informal institutional arrangements to confront uncertainties in the processes of innovation (Peng & Luo, 2000). In general, the extant empirical evidence supports the positive impacts of political ties on innovation (Chen, Han, Wang, Ieromonachou, & Lu, 2021; Tian, Wang, Xie, Jiao, & Jiao, 2019).

Although there is increasing interest in investigating the relationship between political ties and innovation by exploring its intervening and contingent factors (see Appendix A), two crucial aspects of this relationship remain underdeveloped in the political ties-innovation literature. First, different kinds of mechanisms are thought to transform political ties into innovation outcomes (Zhang et al., 2020; Zhang et al., 2019). However, these mechanisms are regarded as homogeneous organizational behaviors that derive from political ties. Rather, these organizational behaviors transform political ties into innovation outcomes based on similar appropriating mechanisms. The theories used in the existing literature (see Appendix A) emphasize the different resources/capabilities acquired from political ties (what is leveraged through social network ties) but neglect the differential use of network connections through political ties (how to leverage social network ties). According to network exploitation and exploration theory, network ties can be leveraged not only by exploiting existing relationships of social ties but also by exploring new relationships through social ties (Lavie & Rosenkopf, 2006). However, in the political ties-innovation literature,

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no study theoretically distinguishes these mechanisms from an exploitation/exploration perspective to empirically compare their roles in transforming political ties into innovation outcomes.

Second, from a relational governance perspective, the effect of political ties is influenced by environmental factors (Sheng et al., 2011). As suggested by Luo (2003), managers dedicate more effort to networking with government officials to alleviate their companies' pressures and pursue opportunities under increased market uncertainty. In the Chinese context, companies resort to relational strategies such as forging personal ties with government officials where formal market institutions are poorly developed (Xin & Pearce, 1996). However, in the political ties-innovation literature, the empirical evidence regarding the moderating effect of environmental factors, such as environmental dynamism, is mixed. For instance, Zhang et al. (2020) find that political ties are rather helpful in facilitating innovation performance amid high levels of environmental dynamism due to the increased importance of resources associated with political ties, which facilitate rapid innovation in dynamic environments, as well as the increased need to exploit new opportunities. In contrast, Tian et al. (2019) show that business-government relations have a weaker impact on innovation when the environment is more dynamic due to a high reluctance to invest in risky innovations and the eroded values of existing resources. Given the salient role played by environmental dynamism, this inconsistency necessitates a deeper understanding of the contingent effects of environmental dynamism on the link between political ties and innovation. The introduction of exploitation/exploration perspectives into political ties-innovation research has great potential to resolve the inconsistent results because the heterogeneous behaviors from exploitation/exploration are preferred and function differently under different uncertain environments (Darr & Kurtzberg, 2000; Gulati, Lavie, & Singh, 2009; Lavie & Rosenkopf, 2006).

Building on network exploitation and exploration theory to further clarify and leverage the potential roles of political ties in innovation, we create and test a mediation and moderation model that integrates the involved underlying mechanisms and contingent factors (see Fig. 1). In particular, we suggest two types of underlying mechanisms (government support and boundary-spanning learning) that explain the exploitative and exploratory leveraging of political ties to increase firm innovativeness. We also introduce market uncertainty, which is the most important component of environmental dynamism, as the contingent factor that moderates the relationships between political ties and underlying mechanisms and the relationships between underlying mechanisms and firm innovativeness. Firm innovativeness refers to a manufacturer's ability to create products that are distinct from existing one and relate to "something new" within the industry (Kunz, Schmitt, & Meyer, 2011). This technology-focused innovation capability drives technology readiness and functions as a crucial enabler of competitive advantage (Parasuraman, 2000). It is a more direct result of technological knowledge than other innovation outcomes such as product innovation performance, as used in Zhang et al. (2019).

2. Theory and hypothesis development

2.1. Network exploitation and exploration of political ties

Since the seminal work of March (1991), both exploration and exploitation have been identified as the two principal activities sought by firms (Lavie, Stettner, & Tushman, 2010; Osiyevskyy, Shirokova, & Ritala, 2020). This point of view has also been given considerable attention in research on network theory (Beckman, Haunschild, & Phillips, 2004; Stadler, Rajwani, & Karaba, 2014). A firm's network could be leveraged in exploitative and exploratory forms. In particular, based on the attribute domain of exploration-exploitation identified in Lavie and Rosenkopf (2006), network exploitation is characterized by the similarities between a firm's partners' attributes, whereas network exploration indicates a deviation in organizational attributes, such as in terms of size or scope, of a firm's partners (Beckman et al., 2004; Huggins, 2010). The stable network associated with the former helps reduce the cost of leveraging existing networks and ensures efficiency (Maurer & Ebers, 2006), while the latter can provide access to diverse new resources, such as technologies and new ideas from partners with different attributes (McEvily & Marcus, 2005; Obstfeld, 2005; Powell, Koput, & Smith-Doerr, 1996).

This study focuses on the network exploitation of political ties in the form of government support and on the network exploration of political ties via boundary-spanning learning. We define government support as the extent to which firms receive support from administrative institutions to limit the adverse impacts of an insufficient institutional infrastructure during a transition period (Guo, Xu, & Jacobs, 2014; Li & Atuahene-Gima, 2001). Government support can be considered as an exploitation of social ties with the government when the associated partners share similar attributes (Lavie & Rosenkopf, 2006). The support could be financial, technical, or take the form of beneficial policies (Li & Atuahene-Gima, 2001). Boundary-spanning learning is defined as the extent to which manufacturers are able to acquire and apply knowledge from universities and research institutes (U&RIs) (Schoenherr & Swink, 2015; Sleep, Bharadwaj, & Lam, 2015). The attributes of U&RIs differ from those of a government and its agencies, indicating companies' network exploration of political ties in terms of boundary-spanning learning (Lavie & Rosenkopf, 2006). U&RIs are the reservoir of original, scientific and innovative knowledge. Boundary-spanning learning can provide firms with access to the novel knowledge generated by U&RIs and with an understanding of the cause-effect relationships among knowledge elements (Ahuja & Katila, 2004; Callahan & Lasry, 2004).

Both network exploitation and exploration are orientated toward

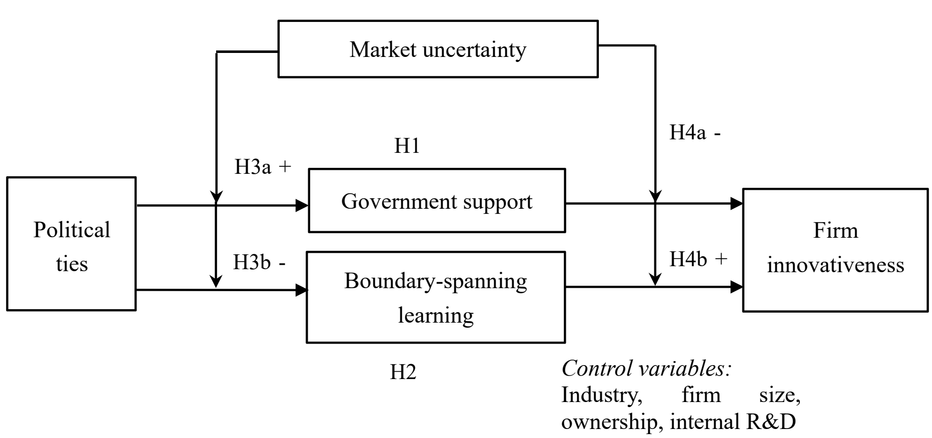

Fig. 1. Conceptual model.

accessing and sourcing knowledge and resources from a firm's partners (Siciliano, Welch, & Feeney, 2018), which influence a firm's innovation capability and performance (Ahuja, 2000). This study focuses on firm innovativeness. Given today's ongoing and rapid market changes, innovativeness is crucial for firms, allowing them to respond positively to market requirements and increase performance (Li, Zhao, & Huo, 2018). According to network exploitation and exploration theory, political ties can be leveraged in both exploitative and exploratory ways to contribute to firm innovativeness. Hence, we propose that government support and boundary-spanning learning help transform political ties into firm innovativeness.

Moreover, uncertainty plays significant roles in influencing a firm's management of network relationships (Beckman et al., 2004; Li & Wang, 2019). Market uncertainty reflects the unpredictability and instability of changing customer preferences in a market (Sheng et al., 2011). Under the condition of high market uncertainty, it is difficult for firms to evaluate the qualities of partners with different characteristics (Gilsing, Nooteboom, Vanhaverbeke, Duysters, & van den Oord, 2008; Obstfeld, 2005). The types of resources firms require are also contingent on market uncertainty (Li & Wang, 2019), which impacts a firm's decision on how to leverage a network relationship. Furthermore, conservative actions, such as stability in interpartner relationships, are favorable for firms, allowing them to counteract or minimize the potential risks caused by market uncertainty (Beckman et al., 2004; Li & Wang, 2019). Accordingly, firms facing high market uncertainty will leverage political ties differently than those facing low market uncertainty. The contributions made by network exploitation and exploration to capability and performance improvement are also contingent on market uncertainty. When market uncertainty is high, frequent changes in customer requirements create a greater need for firms to be innovative to gain competitive advantages (Sheng, Zhou, & Lessassy, 2013). Network exploration can provide access to new knowledge and ideas from outside a firm's domain, promoting the development of legitimately new products. However, since network exploitation is characterized by homogeneity and stability, its value can be eroded by dramatic shifts in customer requirements regarding the creation of new products. Hence, network exploitation and exploration theory indicates that market uncertainty impacts how firms leverage political ties in exploitative and exploratory ways and that the efficiency of exploitative and exploratory network leveraging influences firm innovativeness differently. Therefore, we argue that market uncertainty impacts not only the relationships between political ties and government support or boundary-spanning learning but also the effects of government support or boundary-spanning learning on firm innovativeness.

2.2. Mediating effects of government support and boundary-spanning learning

Political ties help firms obtain government support. Firms with numerous political ties are given high priority by government bodies regarding resource allocations (Li & Zhang, 2007). The established interactive mechanisms and experiences between firms and a government also render firms capable of influencing their political environment to obtain institutional resources (Guo et al., 2014; Zhang et al., 2019). Political ties can provide a manufacturer with political legitimacy, facilitating its acquisition of resources, such as financial support and technical information (Guo et al., 2014; Zhang, Jiang, Wu, & Li, 2019). Connections with government officials also help reduce any unfair or opportunistic behaviors used by competitors to acquire resources from a government; such connections can also provide a firm with influence over regulatory policies (Li & Zhang, 2007). Moreover, given similar attributes among the embedded relationships of political ties and government support, the risk and cost of exploiting existing relationships are relatively low (Beckman et al., 2004; Li & Wang, 2019). Accordingly, firms are motivated to leverage political ties to obtain support from the government due to the benefits associated with government support.

Government support can also contribute to firm innovativeness through an enhanced motivation to invest in innovation and improved success in innovation. Government support in the forms of technical information and financial support can relieve concerns about the risk for innovation (Zhang, Wang, Zhao, & Zhang, 2017). Moreover, the signaling or certification effects of government R&D subsidies can motivate third-party investors to support the development of new products (Wu, 2017). Policy and regulatory support from a government can help to reduce uncertainty in innovation (Peng, 2002) and to protect innovation outcomes, such as intellectual capital. This, in turn, can increase a firm's willingness to invest in new product development (Wang, Sutherland, Ning, Wang, & Pan, 2018; Wu, 2017). Moreover, according to the resource-based view, government support provides rare and valuable resources for firms' competitive advantage in innovativeness (Barney, 1991). Previous research also suggests that government support enhances both product and process innovation (Shu, Wang, Gao, & Liu, 2015; Zhang et al., 2017).

Accordingly, we propose that government support will mediate the effects of political ties on firm innovativeness. Although political ties provide vital opportunities for a firm's new product development, close interpersonal relationships are not enough to generate a new product (Hult, Ketchen, Cavusgil, & Calantone, 2006; Zhang, Zhao, & Lyles, 2018). Firm innovativeness tends to be more immediately impacted by government support due to the associated resources. Thus, we propose the following hypothesis:

Hypothesis 1. *Government support mediates the relationship between political ties and firm innovativeness.*

Political ties can also promote firm innovativeness through boundary-spanning learning. Political ties can be used to explore resources from U&RIs, which have different attributes than a government and its agencies. Political ties offer a firm a better understanding of government priorities (Li & Zhang, 2007), while the collaboration between firms and U&RIs is emphasized and encouraged by the Chinese government. Firms with political ties can also influence policies and regulations to favor or facilitate their collaboration with U&RIs (Zhang, 2006), providing manufacturers with sufficient access to acquire and apply U&RI knowledge (Wang, Jiang, Yuan, & Yi, 2013). Moreover, U&RIs in China are dependent on the government for R&D support (Gao, Xu, & Yang, 2008). The connection that certain firms with political ties and U&RIs share with China's government provides those firms with opportunities for collaboration with U&RIs. Political ties can also provide external legitimacy to a firm (Zimmerman & Zeitz, 2002), reducing its opportunistic behaviors regarding knowledge transfer from U&RIs. Firms with many political ties are typically more capable of realizing boundary-spanning learning. Moreover, a firm's leveraging of political ties to facilitate boundary-spanning learning is also critical to renew its knowledge of innovation, given the current climate of intense competition, short product lifecycles, and U&RIs' pioneering technological knowledge.

Organizational boundary spanning is essential for a firm's innovation, given that nearly all firms have insufficient internal knowledge resources, especially in dynamic environments (Gumusluoglu & Ilsev, 2009). Boundary-spanning learning helps increase firm innovativeness. First, firms can access and acquire novel knowledge, especially scientific inventions and breakthrough technologies (Hemmert, 2004), from U&RIs to readily leverage expert knowledge for new product development (Veugelers & Cassiman, 2005). Second, knowledge from U&RIs can allow manufacturers to better understand the cause-effect relationships between relevant knowledge elements (Ahuja & Katila, 2004). Such an understanding facilitates new product development and thus contributes to firm innovativeness. The importance of boundary-spanning learning for innovation, in the form of collaboration with U&RIs, has been emphasized in the literature (Un & Asakawa, 2015; Un, Cuervocazurra, & Asakawa, 2010).

Accordingly, we argue that boundary-spanning learning will mediate the effects of political ties on firm innovativeness. First, resources and knowledge play a rather immediate role in influencing a firm's ability to introduce new products (Wang et al., 2013). Second, firms with strong political ties might be restricted by complying with government requirements when exploring new opportunities. Boundary-spanning learning can help firms overcome this issue by providing access to novel knowledge for innovation, which, therefore, helps firms realize the benefits of their political ties, fostering firm innovativeness. Thus, we propose the following hypothesis:

Hypothesis 2. *Boundary-spanning learning mediates the relationship between political ties and firm innovativeness.*

2.3. Moderating effects of market uncertainty

High market uncertainty indicates unpredictability and instability due to changing customer preferences in a market (Sheng et al., 2011). Market uncertainty can substantially impact the effects of political ties on government support and boundary-spanning learning. We suggest that market uncertainty positively moderates the impact of political ties on government support but negatively moderates the impact of political ties on boundary-spanning learning.

We propose that when a market has high uncertainty, firms are more motivated to commit their resources to seek conservative actions, such as emphasizing exploitation to generate more reliable returns and avoid another layer of uncertainty, but are less likely to take progressive actions, such as exploration (Beckman et al., 2004; Li & Wang, 2019). Government support not only indicates a firm's exploitation of social connections with a government but is also helpful in reducing the challenges caused by uncertainty. Guo et al. (2014) suggest that government support can be utilized to make use of environmental uncertainties. Leveraging political ties to source government support is thus more favorable in uncertain markets. However, given high market uncertainty and a long period associated with a commercialization of scientific knowledge from U&RIs (Ahuja & Katila, 2004), in addition to the inherent difficulties in assessing the qualities of U&RIs with attributes that differ from a government's, firms can be reluctant to leverage their social connections with a government to obtain boundary-spanning learning from U&RIs. Firms with strong political ties also tend to be less tolerant of risk (Sheng et al., 2011). Thus, although political ties are of greater importance, they are less likely to be used to facilitate boundary-spanning learning from U&RIs when a market is highly uncertain.

In contrast, when market uncertainty is low, firms with political ties are in less need of and have less incentive to seek government support, as market changes are more predictable. In a stable market, exploring the knowledge from U&RIs with different attributes can help firms relieve the risk of overemphasizing connections with a government and help them access new knowledge for innovation generation to apply in their current competitive environment. The benefits from boundary-spanning learning are more attractive, and firms with political ties are therefore more likely to be motivated to invest in boundary-spanning learning. Thus, we propose the following hypotheses:

Hypothesis 3a. *The effect of political ties on government support is higher when the level of market uncertainty is high.*

Hypothesis 3b. *The effect of political ties on boundary-spanning learning is lower when the level of market uncertainty is high.*

Furthermore, market uncertainty also impacts the relationship between government support or boundary-spanning learning and firm innovativeness. Hence, we argue that market uncertainty will negatively moderate the relationship between government support and firm innovativeness but positively moderate the relationship between boundary-spanning learning and firm innovativeness. When the level of market uncertainty is high, products with radical new features appeal to consumers and function as a source for competitive advantage (Sheng et al., 2013; Zhang et al., 2020). This requires firms to explore knowledge sources that could provide diverse knowledge and enable the development of radical product innovations. Boundary-spanning learning provides variations in technologies, supporting the achievement of radical breakthrough innovations (Callahan & Lasry, 2004). However, in this context, the value of government support for firm innovativeness is low. In an unpredictable market, firms with high government support are overembedded in their relationships with a government, restricting their capability to source novel knowledge and their flexibility to identify and respond to new customer demands. The information provided by a government, for example, industrial or regional economic output data, can easily become outdated in unstable markets (Blind, Petersen, & Riillo, 2017; Sheng et al., 2011), constraining the efficiency of government support that fosters firm innovativeness. In addition, the resources associated with government support are often short-term oriented (Sheng et al., 2011), which is less attractive for firms to innovate amid changing and unpredictable customer requirements.

When customer demand is predictable and stable, financial support and technological information facilitate the introduction of new products. The importance of boundary-spanning learning is not prominent in a stable market due to integrated considerations of its uncertainty and the limited requirements for novel knowledge. Thus, we propose the following hypotheses:

Hypothesis 4a. *The effect of government support on firm innovativeness is lower when the level of market uncertainty is high.*

Hypothesis 4b. *The effect of boundary-spanning learning on firm innovativeness is higher when the level of market uncertainty is high.*

3. Research methodology

3.1. Questionnaire design

A research questionnaire was designed to measure the political ties of senior managers along with their firms' government support, boundary-spanning learning, firm innovativeness, and degree of market uncertainty. It also included certain company characteristics. All items were rated on a 7-point Likert scale. *Political ties* were measured by five items adapted from Peng and Luo (2000) and Li, Poppo, and Zhou (2008) to indicate the extent to which top managers have built personal connections with leaders and officials of the Chinese government and its agencies. Following Li and Atuahene-Gima (2001), *government support* was measured with four items that reflect the extent to which manufacturers have obtained support from administrative institutions (Guo et al., 2014). Following van der Bij, Song, and Weggeman (2003) and Song, van der Bij, and Weggeman (2005), *boundary-spanning learning* was measured using five items to indicate a firm's ability to acquire and apply knowledge sourced from U&RIs. *Firm innovativeness* was measured by three items adapted from Parasuraman (2000) and Tellis, Prabhu, and Chandy (2009) to reflect a firm's ability to create products that are distinct from existing ones and refers to "something new" within the industry (Kunz et al., 2011). *Market uncertainty* was measured by three items adapted from Chen and Paulraj (2004) to indicate the unpredictability and instability of customer preferences within a market.

In addition, given the potential impacts of firm size, ownership, industry and internal R&D on the dependent variables, we also identified and controlled for their impacts in this study. In particular, firm size was measured by the number of employees in a firm. We used dummy variables to control for the impacts of ownership and industry. Foreign firms and firms in an industry of new materials were selected as the baseline. Internal R&D was measured by average research and development budget, as a percentage of total firm sales.

3.2. Sampling and data collection

We conducted this study in China for the following two reasons: First, social connections with the government play more significant and unique roles in influencing ways of doing business in China than in other emerging economies, such as India (Zhang et al., 2019). Second, the existing literature on political ties is largely rooted in China (Jean, Sinkovics, & Zagelmeyer, 2018; Zhang et al., 2020). Our use of a similar research setting makes the connections and comparisons of this study with existing ones more convincing.

We collected the data through the following steps: First, 15 Chinese manufacturing companies were selected from three special economic zones in China (i.e., the Pearl River Delta, the Yangtze River Delta, and the Bohai Rim Economic Circle) to conduct a pilot test of the questionnaire. Then, using the catalog provided by the National Bureau of Statistics of the People's Republic of China, 2379 manufacturing companies were randomly selected from the target industries in the three regions. Second, a key information provider for each company was identified. The providers were general managers, directors, senior R&D managers, operations or manufacturing managers, and supply chain managers. They needed to have personal connections with government officials and to be knowledgeable about their company's knowledge management and innovation. Third, the research team hired a professional market research company to collect data. To identify potential respondents and invite them to participate in the survey, the market research company called each target company. A total of 2061 companies in the selected sample were unwilling to participate in the survey or were excluded because of a lack of correct contact information. The remaining 318 companies were investigated by the market research company through on-site visits, and 300 complete questionnaires were collected: a response rate of 12.6% (300/2379). Excluding companies without boundary-spanning learning generated a final sample of 262 firms. The basic characteristics of the 262 samples, based on firm size, ownership, industry and internal R&D investment, are shown in Table 1.

3.3. Reliability and validity

We compared the first and last 30 respondents to assess nonresponse bias (Collier & Bienstock, 2007). The results of a *t*-test showed that there is no significant difference between the two batches of data samples regarding all the key constructs used in this study at a confidence level of 95%. This indicates that the problem of nonresponse bias in this study is not serious.

The survey instrument's structural reliability was assessed via Cronbach's α and composite reliability (CR). The values of Cronbach's

Table 1
Basic characteristics of the samples (n = 262).

Variables	Category	Number	Percentage
Firm size	101–200	56	21.4
	201–300	53	20.2
	301–500	60	22.9
	501–1000	43	16.4
	More than 1000	50	19.1
Ownership	State-owned	89	34
	Privately owned	94	35.9
	Joint venture	41	15.6
	Foreign investment	38	14.5
Industry type	Biology & pharmaceuticals	18	6.9
	Computers & telecommunications equipment	25	9.5
	Chemicals	48	18.3
	Medical equipment	25	9.5
	Electronics & electrical equipment	43	16.4
	Industrial machinery	44	16.8
	Transportation equipment	34	13
	New materials	25	9.5
Internal R&D	0–0.25%	5	1.9
	0.26–0.50%	22	8.4
	0.51–0.75%	7	2.7
	0.76–1%	16	6.1
	1.1–2%	39	14.9
	2.1–4%	127	48.5
	>4%	46	17.6

Table 2
Scale reliability and validity.

Measurement	Factor loading
Political ties: *Please circle the number that best describes the extent to which top managers at your firm have utilized personal ties, networks, and connections during the past three years with the following departments (1 = very little, 7 = very extensively)* (Cronbach's α = 0.87; CR = 0.87; AVE = 0.57).	
1. Political leaders in various levels of the government	0.81 (0.09[a],12.26[b])
2. Officials in industrial bureaus	0.73(0.08,10.96)
3. Officials in regulatory and supporting organizations, such as science & technology parks, tax/commercial administration bureaus, state banks, or small and medium enterprise administration bureaus	0.80(0.08,12.20)
4. Officials in science and technology bureaus	0.71(0.09,10.81)
5. Officials in state authorities that make laws, regulations, and policies	0.73
Government support: *Please indicate the extent to which, in the last three years, the government and its agencies have (1 = very little, 7 = very extensive)* (Cronbach's α = 0.85; CR = 0.85; AVE = 0.59):	
1. Implemented policies and programs that have been beneficial to your company's operations.	0.72(0.08,10.21)
2. Provided needed technology information and technical support to your company.	0.84(0.09,11.54)
3. Played a significant role in providing financial support for your company.	0.81(0.09,12.01)
4. Helped your company to obtain licenses for imports of technological, manufacturing or other equipment.	0.70
Boundary-spanning learning: *Please indicate your degree of agreement with the following statements assessing the roles of major universities and research institute in your company's innovation (1 = very little, 7 = very extensive)* (Cronbach's α = 0.92; CR = 0.92; AVE = 0.69):	
1. We are able to obtain a tremendous amount of our technological knowledge from universities/research institutes.	0.85(0.08,14.40)
2. We rapidly respond to technological changes in our industry by applying what we know from universities/research institutes.	0.88(0.07,14.95)
3. As soon as we acquire new knowledge from universities/ research institutes, we try to find applications for it.	0.82(0.07,14.13)
4. University/research institute's technological knowledge has enriched the basic understanding of our innovation activities.	0.84(0.07,14.83)
5. University/research institute's technological knowledge helps us to identify new aspects of innovation activities that would otherwise go unnoticed.	0.77
Market uncertainty *Please indicate your degree of agreement with the following statements on environmental uncertainty (1 = very little, 7 = very extensive)* (Cronbach's α = 0.85; CR = 0.83; AVE = 0.65):	
1. It is difficult to predict changes in the market.	0.88(0.14,10.24)
2. The volume and/or composition of demand is difficult to predict.	0.90(0.14,10.29)
3. Customer requirements for products can change dramatically.	0.61
Firm innovativeness: *Please indicate your degree of agreement with the following statements describing your company's product innovation (1 = very little, 7 = very extensive)* (Cronbach's α = 0.77; CR = 0.77; AVE = 0.52):	
1. We are the first within the industry to introduce new products.	0.73(0.10,9.20)
2. We keep up with the latest product developments.	0.69(0.09,8.82)
3. We frequently introduce products that are radically different from established products in the industry.	0.75
Financial performance: *Your firm's overall performance compared with a major competitor over the past year regarding (1 = far worse, 7 = far better)* (Cronbach's α = 0.849; CR = 0.85; AVE = 0.59):	
1. Total sales of product and service	0.85(0.13, 10.59)
2. Profit	0.71(0.11, 9.90)
3. Market share	0.84(0.13, 10.74)
4. Sales growth rate	0.65

Notes: a: standard error; b: t-value; CR = composite reliability; AVE = average variance extracted.

alpha ranged from 0.77 to 0.92, and the values of CR ranged from 0.77 to 0.92 (as shown in Table 2); both are above the recommended threshold of 0.70.

In terms of validity, first, a confirmatory factor analysis (CFA) was used to assess convergence validity (O'Leary-Kelly & Vokurka, 1998). Each measurement item was associated with its corresponding construct, and the covariance between constructs was freely estimated. The model fitting indices were $\chi^2(160) = 289.055$, $\chi^2/df = 1.807$, the goodness-of-fit index (GFI) was 0.900, the normed fit index (NFI) was 0.902, the comparative fit index (CFI) was 0.953, the Tucker Lewis index (TLI) was 0.944, and the root mean square error of approximation (RMSEA) was 0.056. These results are superior to the thresholds recommended by Hu and Bentler (1999), indicating convergence. In addition, as shown in Table 2, all factor loadings are>0.50 (range 0.60 to 0.84), the t values are>2.0 (Koufteros, Cheng, & Lai, 2007), and the coefficient of each item exceeds twice its standard error (SE) (Anderson & Gerbing, 1988). Furthermore, the estimates for the average variance extracted (AVE) are higher than 0.50 for our constructs (Fornell & Larcker, 1981). Therefore, our constructs have convergent validity.

Second, discriminant validity was evaluated by comparing the square root of each construct's AVE with that of each other construct (Fornell & Larcker, 1981). Table 3 shows the mean and standard deviation of each construct and the two-two correlation coefficient, which indicates that the square root of AVE is greater than the correlation coefficient of any one of the two constructs; therefore, the discriminant validity of the model is established.

4. Analysis and results

4.1. Descriptive statistics and correlations

Table 3 shows the mean, standard deviation and correlation coefficient matrix of the main variables of this study. The table demonstrates that political ties, government support, boundary-spanning learning and firm innovativeness are significantly positively correlated with each other. A review of the correlations among the independent variables suggests that multicollinearity is not a major concern, as confirmed by the variance of inflation (VIF), which ranges from 1.01 to 1.50 (Hair, Black, Babin, Anderson, & Tatham, 1998).

4.2. Controlling for common method bias

In this study, both *ex ante* and *ex post* approaches were applied to control for the possible effects of common method bias (CMB). In the design stage, we used techniques, such as an anonymous measurement and reverse measurement of some items. In addition, we selected the key information provider for each company to complete the questionnaire accurately. We knew the name of each company and ensured the strictest confidentiality to confirm the identities of the respondents.

After the data were collected, we conducted three statistical tests to measure the potential effects of common method bias. First, we used Harman's single-factor method for statistical tests. The unrotated factor analysis showed that the KMO value was 0.848, the chi-square value was 2956.390, the significance level was 0.000, and the single-factor maximum variance contribution rate was 28.737%. Therefore, the CMB of this study is within an acceptable range. Second, a CFA model, in which the common method was treated as a single-factor loading on all items, was constructed (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). The results showed that $\chi^2/df = 10.29$, CFI = 0.425, TLI = 0.357, RMSEA = 0.189, which indicates poor model fitness (Hu & Bentler, 1999). Finally, a common factor loading on all items was added to the CFA model to estimate the amount of variance from each item that can be attributed to the common method (Podsakoff et al., 2003). The results showed that $\Delta\chi^2/df = 0.28$, ΔCFI = 0.02, ΔTLI = 0.02, ΔRMSEA = 0.011, ΔSRMR = 0.0103, indicating that the model was not improved significantly by adding a common factor. Therefore, the problem of CMB in this study is not serious.

4.3. Hypothetical test

PROCESS analysis, a bootstrapping technique (Hayes, 2012) that allows multiple mediators in model testing (Zhao, Lynch, & Chen, 2010), was used to test our hypotheses. The results of the simple mediation analysis are summarized in Table 5. H1 and H2 state that government support and boundary-spanning learning mediate the relationship between political ties and firm innovativeness. As Table 4 shows, the direct effect of political ties on firm innovativeness was not significant when the two mediators were controlled for ($p = 0.369$). However, based on the bootstrapping results, the indirect effects on firm innovativeness from government support ($b = 0.144$, SE = 0.044, confidence interval (CI) of 95% is [0.065, 0.239]) and from boundary-spanning learning ($b = 0.080$, SE = 0.029, CI of 95% is [0.032, 0.146]) were both significant at the 0.05 level. Thus, H1 and H2 are supported. Furthermore, government support plays a more important role than boundary-spanning learning in how political ties impact firm innovativeness.

In addition, we employed an analysis of moderated mediation. To reduce the potential problem of multicollinearity, both contingent variables were mean-centered (Aiken, West, & Reno, 1991). Table 5 presents the estimation results regarding the moderating roles played by market uncertainty in the process of how political ties influence firm innovativeness. Hypothesis 3a proposes that the positive relationship between political ties and government support will be stronger when the level of market uncertainty is high. This hypothesis is not supported because the coefficient for the interaction of political ties and market

Table 4
Results of the simple mediation analysis.

Direct Effect			
	Effect	SE	95% CI
Political ties → firm innovativeness	−0.064 (p = 0.446)	0.071	[-0.204, 0.076]
Indirect Effect			
	Effect	Boot SE	95% CI
H1: Political ties → government support → firm innovativeness	0.144	0.044	[0.065, 0.239]
H2: Political ties → boundary-spanning learning → firm innovativeness	0.080	0.029	[0.032, 0.146]

Notes: SE = standard error; CI = confidence interval.

Table 3
Mean, standard deviation and correlation coefficient matrix.

Variable	Mean	S.D.	(1)	(2)	(3)	(4)	(5)
(1) Political ties	5.12	1.02	**0.75**				
(2) Government support	4.59	1.13	0.51***	**0.79**			
(3) Boundary-spanning learning	5.11	1.14	0.31***	0.28***	**0.83**		
(4) Market uncertainty	4.17	1.31	−0.07	−0.02	0.07	**0.81**	
(5) Firm innovativeness	5.31	1.00	0.17*	0.34***	0.31***	−0.02	**0.72**

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; the bold on the diagonal is the square root of the average variation extraction.

Table 5
Results of the Moderated Mediation.

Variable	Model 1 Government support	Model 2 Boundary-spanning learning	Model 3 Firm innovativeness
Constant	0.318	0.251	−0.093
Political ties (PT)	0.510***	0.340***	−0.054
Market uncertainty (MU)	0.026	0.108+	0.0002
H3a, H3b: PT × MU	−0.005	−0.179**	
Government support (GS)			0.285***
Boundary-spanning learning (BSL)			0.237***
H4a: GS × MU			−0.112+
H4b: BSL × MU			0.128*
State-owned	−0.503**	−0.126	−0.375
Privately owned	−0.111	−0.050	−0.312+
Joint venture	−0.240	−0.524*	−0.215
Biology & Pharmaceuticals	−0.243	−0.550+	−0.317
Computer & Telecommunication equipment	−0.318	−0.494+	−0.218
Chemicals	−0.165	−0.493*	−0.094
Medical equipment	−0.293	−0.188	−0.108
Electronics & electrical equipment	−0.112	−0.328	−0.321
Industry machinery	0.054	−0.251	−0.072
Transportation equipment	−0.093	0.012	−0.298
Firm size	0.063	−0.027	0.016
Internal R&D	−0.035	0.051	0.086*
R^2	0.309	0.183	0.230
F	7.330	3.675	4.025

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

uncertainty is not significant (b = -0.005, p > 0.05). Hypothesis 3b, which proposes that the positive relationship between political ties and boundary-spanning learning will be lower when the level of market uncertainty is high, is supported: the coefficient for the interaction of political ties and market uncertainty is negative and significant (b = -0.179, p < 0.01).

We plot this interaction effect in Fig. 2 by following the procedure described by Aiken et al. (1991) to facilitate interpretations of this finding. Fig. 2 shows that political ties have a higher positive relationship with boundary-spanning learning when the level of market uncertainty is low.

Hypothesis 4a states that government support has a weaker positive relationship with firm innovativeness when the level of market uncertainty is high. This hypothesis is supported; the coefficient for the interaction of government support and market uncertainty is negative and significant (b = -0.112, p < 0.1). Hypothesis 4b, which proposes that the positive relationship between boundary-spanning learning and firm innovativeness will be stronger when the level of market uncertainty is high, is also supported. The coefficient for the interaction of boundary-spanning learning and market uncertainty is positive and significant (b = 0.128, p < 0.05).

Following a similar procedure to that described above, we plot the moderating effects of market uncertainty in Fig. 3 and Fig. 4. As shown in Fig. 3, the positive relationship between government support and firm innovativeness is weaker when the level of market uncertainty is high.

As shown in Fig. 4, the positive relationship between boundary-spanning learning and firm innovativeness is stronger when the level of market uncertainty is high.

In addition, we adopt the approach used in Zhang et al. (2020) to analyze the conditional indirect effects of government support and boundary-spanning learning on high market uncertainty (mean + 1 SD) and low market uncertainty (mean-1 SD). The results show that when market uncertainty is high, the indirect effect of political ties on firm innovativeness through government support becomes weaker (indirect effect $b = 0.087$, SE = 0.050, CI of 95% is [0.001, 0.199]), and the indirect effect of political ties on firm innovativeness through boundary-spanning learning is not significant (indirect effect $b = 0.059$, SE = 0.038, CI of 95% is [-0.001, 0.156]). When market uncertainty is low, the indirect effect of political ties on firm innovativeness through government support is stronger (indirect effect $b = 0.205$, SE = 0.061, CI of 95% is [0.101, 0.344]), and the indirect effect of political ties on firm innovativeness through boundary-spanning learning is, again, not significant (indirect effect $b = 0.057$, SE = 0.052, CI of 95% is [-0.048, 0.158]).

4.4. *Endogeneity check*

We conducted a post hoc analysis to address possible endogeneity issues. First, political ties can sometimes be explained by other environmental factors. Zhou, Li, Sheng, and Shao (2014) posit that the formation of political ties by senior executives is affected not only by industry factors, such as market uncertainty and competition intensity, but also by institutional environments, such as legal effectiveness. To address this issue, this study used as many control variables that affect both the independent and dependent variables in the model as possible (Lu, Ding, Peng, & Chuang, 2018). Dysfunctional competition, which refers to unfair, opportunistic or even unlawful market practices within a firm's industry, represents the degree of legal effectiveness (Li & Zhang, 2007). Since market uncertainty has been included in our model, we have used competition intensity and dysfunctional competition as control variables that are included in the regression model. After testing,

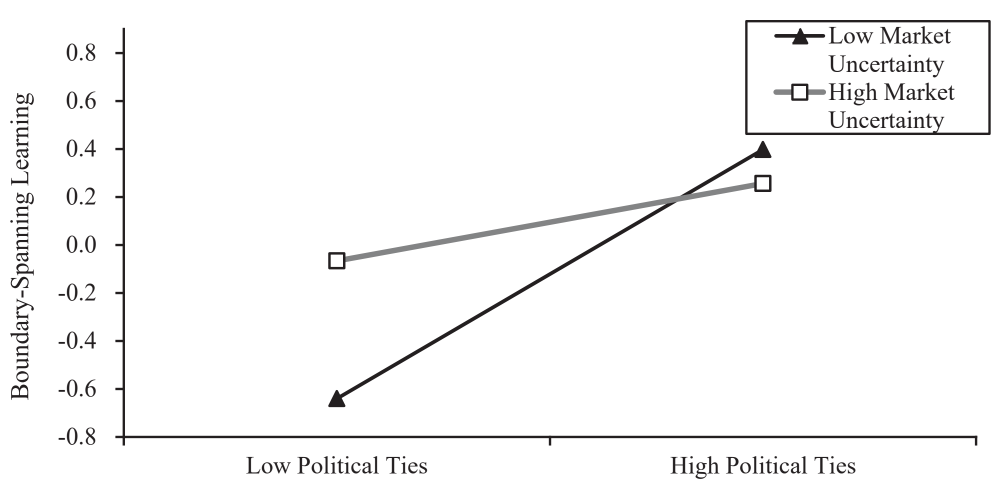

Fig. 2. Moderation of market uncertainty on the relationship between political ties and boundary-spanning learning.

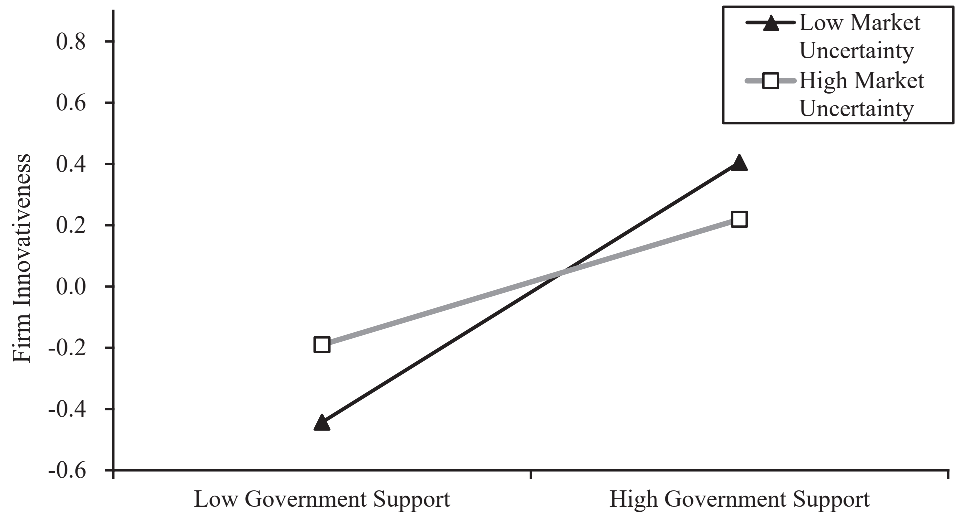

Fig. 3. Moderation of market uncertainty on the relationship between government support and firm innovativeness.

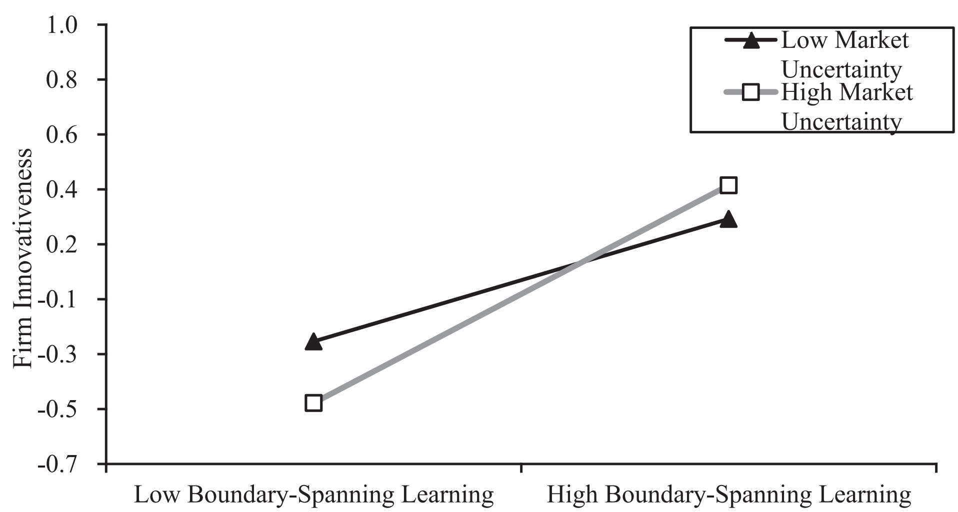

Fig. 4. Moderation of market uncertainty on the relationship between boundary-spanning learning and firm innovativeness.

the regression results were still significant.

Second, political ties and government support can be affected by the business performance of a firm. Reverse causality may affect the mediating effects of government support on the relationship of political ties with firm innovativeness. Fan, Wong, and Zhang (2007) indicate that good business performance will attract both the attention of China's government and more support from it, which can help a firm accumulate additional political connections. To overcome this potential problem, following the logic of Fan et al. (2007), the study sample was divided into firms with "low" and "high" scores according to two performance indicators to identify whether the original relationship held for both subgroups. The two performance indicators were financial performance

Table 6
Endogenous test.

Variable	Government support				Firm innovativeness			
	FP Low	FP High	NP Low	NP High	FP Low	FP High	NP Low	NP High
Constant	1.124*	0.549	0.266	2.101^{***}	−0.509	0.477	0.035	0.217
Political ties	0.406^{***}	0.494^{***}	0.409^{***}	0.334^{**}				
Government support					0.418^{***}	0.262*	0.399^{***}	0.318*
Firm size	0.049	−0.019	−0.072	−0.026	−0.024	0.047	0.144^{+}	−0.049
Internal R&D	−0.088	−0.085	−0.047	-0.188^{**}	0.135*	0.013	0.087	0.051
Ownership	Control	Control	Control	Control	Control	Control	Control	Control
Industry	Control	Control	Control	Control	Control	Control	Control	Control
Observations	107	98	100	95	107	98	100	95
Adjusted R^2	0.22	0.081	0.174	0.252	0.154	−0.03	0.159	0.069
F	3.141	1.66	2.492	3.257	2.375	0.783	2.341	1.498

Notes: $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; 4 and 8 dummy variables were generated for ownership and industry, respectively. To save space, the table does not list specific regression coefficients. Full results are available from the authors upon request.

(FP) and the number of new patents (NP). As shown in Table 2, FP was a subjective variable, measured with four, 7-point Likert-scale items (Cronbach's $\alpha = 0.849$; CR = 0.85; AVE = 0.59). NP was an objective variable, measured with firms' total number of invention patents from 2000 to 2012 (before the year of questionnaire collection). The sample was then divided into "low" and "high" groups according to the median, after which whether the political ties of executives significantly impact government support and whether government support significantly impacts firm innovativeness were tested for each indicator. The results in Table 6 show that the regression coefficients of executive political connections to government support and the regression coefficients of government support to firm innovativeness are significant for the two performance indicators.

Finally, NP was not correlated with political ties ($b = 0.007$, $p > 0.1$) or government support ($b = 0.055$, $p > 0.1$), which confirms that reverse causality does not significantly affect the relationship between political ties, government support and firm innovativeness. Accordingly, our results suggest that the effects of exogenous issues on the model are not serious.

5. Discussion

5.1. Theoretical contributions

This study makes three contributions to the existing literature. First, we contribute to managerial ties research by distinguishing two types of underlying mechanisms that transform political ties into innovation performance. In contrast to the extant studies, by introducing network exploitation and exploration theory, this study increases the existing knowledge about the transformative mechanisms connecting political ties and innovation and empirically reveals how political ties can be leveraged in both exploratory and exploitative ways. Accordingly, the processes through which the values of political ties are realized are identified and enriched. That is, political ties can be leveraged in exploitative ways by reinforcing firms' existing relationships with governments and in exploratory ways when firms develop new relationships with partners, such as universities and research institutions; these relationships have different attributes than firms' existing relationships with governments. This distinction is important because these two approaches capture two distinct facets of political ties and provide access to different resources. Taken together, our theoretical logic and empirical findings enrich the current understanding of how political ties contribute to firm innovativeness (Guo et al., 2014).

Specifically, the findings demonstrate that boundary-spanning learning is a new mechanism for understanding the transformation of political ties into innovation performance. The important role that we show boundary-spanning learning from U&RIs plays in this transformation also extends the current research, which primarily focuses on political and business stakeholders, such as suppliers, customers and competitors (Li et al., 2018; Zhang et al., 2020). The findings contribute to the relationship-learning view of firm strategy (Newton, Tsarenko, Ferraro, & Sands, 2015). Political ties, as firms' strategic options, not only directly generate government support but also indirectly facilitate relationship learning activities.

Second, we enrich the network exploitation and exploration literature by conceptualizing network exploitation and exploration in the context of social ties. Our findings treat political ties as strategic means for firms to confront uncertain environments in both exploitative and exploratory ways. The existing literature, however, has focused on the exploration and exploitation of different forms of strategic alliances (Lavie & Rosenkopf, 2006). Few empirical studies have investigated exploration and exploitation behaviors in the context of social ties. Compared with strategic alliances, social ties are informal and loosely coupled connections between firms. By recognizing the particular importance of network exploitation/exploration theory for social ties research, this study is the first to empirically predict and demonstrate two diverse leveraging behaviors in the context of political ties in the transition economy of China.

Third, we shed light on the complex process that binds political ties to firm innovation. Our results help to explain the previously inconclusive findings regarding the moderating effect of environmental dynamism by simultaneously considering the exploitative and exploratory mechanisms that leverage political ties to facilitate firm innovativeness under different environmental conditions. This study discloses the different influences of market uncertainty on the relationships between political ties, government support, boundary-spanning learning, and firm innovativeness. Our findings highlight how and why political ties impact government support and boundary-spanning learning differently across high- and low-uncertainty markets. While the existing literature has investigated the contingent impacts of an external environment on the link between political ties and innovation, our findings disclose some rather detailed effects of market uncertainty on the integration of network ties and multiple mechanisms for firm innovation. Hence, this study contributes to the literature on political ties and innovation by explaining how market uncertainty exerts influences on the complex process linking political ties to firm innovativeness.

To enrich our findings further, the role of government support requires further investigation. Our proposed positive moderating effect of market uncertainty on the link between political ties and government support is not supported. One possible reason could be that the resource benefits of government support for a firm's operations are attractive enough for a firm to exploit its political ties to gain government support even when a market is stable and predictable. Another reason could be that although exploiting political ties to source government support is of relatively low uncertainty, the potential benefit of any information or knowledge provided by a government fades quickly in a market with changing customer requirements (Sheng et al., 2011). This can restrict a firm's motivation to utilize political ties for government support. Accordingly, the relationship between a firm's political ties and government support is not sensitive to the level of market uncertainty.

5.2. Managerial implications

Our findings provide meaningful insights for manufacturers on how to leverage political ties to enhance their innovativeness. Based on our findings, senior managers of manufacturers in China should still focus on the importance of political ties in firms' operations and establish and maintain good personal connections with political leaders and government officials. While political ties provide opportunities to access external resources, it is vital for firms in China to exploit political ties to seek government support and explore political ties to acquire and apply the knowledge they gain from U&RIs to obtain innovation benefits from political ties. Meanwhile, managers should also consider the environmental conditions of their firms when leveraging political ties to enhance firm innovativeness. In particular, when a market is stable and predictable, an emphasis on using political ties to seek government support to obtain resources for firm innovativeness is advantageous. However, the adverse moderating effects of market uncertainty on the link between political ties and boundary-spanning learning, and on the link between boundary-spanning learning and firm innovativeness, underscore the challenge that a "capability gap" presents to firms. In an unstable market, exploring political ties of boundary-spanning learning is less attractive, but boundary-spanning learning will foster firm innovativeness. In this situation, managers should create more political ties to enhance boundary-spanning learning, the real value of which may not be immediately evident. A more balanced view regarding how firms leverage political ties to promote government support and boundary-spanning learning for firm innovativeness should be emphasized by managers. In addition, from a government official's perspective, establishing connections with top managers is also a potential means to improve a firm's innovation capability.

5.3. Limitations and future research

Although this study has made several significant theoretical contributions that have several managerial implications, there are still some limitations that necessitate further investigation. First, in this study, we used government support and boundary-spanning learning to represent the exploitative and exploratory leveraging of political ties. The roles of other exploitative and exploratory behaviors such as institutional entrepreneurial opportunity recognition (Guo et al., 2014) and information acquisition from supply chain partners (e.g., foreign customers) (Lu, Zhou, Bruton, & Li, 2010) in realizing the potential for political ties to enhance firms' innovativeness could also be explored in future research. Second, this study considers only the moderating role played by market uncertainty. Although it is one of most vital environmental factors, there are other environmental factors, such as technological uncertainty and market competition, that impact firms' decisions on how to leverage political ties for firm innovativeness. Further research is needed to understand the contingent effects of political ties on influencing innovation. Third, this study was based in China, a specific research context, which restricts the generalization of its findings. Other research has suggested that transitional economies also have distinct characteristics that differentiate resource allocation and utilization in innovation (Zhang et al., 2019). Future studies could investigate and compare this research framework across countries.

CRediT authorship contribution statement

Zhiqiang Wang: Conceptualization, Writing – original draft, Writing – review & editing. **Xiaoli Chen:** Writing – original draft, Methodology, Formal analysis. **Shanshan Zhang:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Ying Yin:** Project administration, Data curation. **Xiande Zhao:** Project administration, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. . Summary of the literature regarding the relationship between political ties and innovation

Literature	Theory	Dependent variable	Mediators	Moderators	Related findings
Chen et al. (2021)	Social capital theory, resource dependency theory, institutional theory	Innovation performance	–	Environmental inefficiency, survival pressure	Political ties (*guanxi* political ties and transactional political ties) positively impact innovation. Environmental inefficiency and survival pressure impact these relationships.
Gao et al. (2017)	Social network theory, institution theory	Product innovation	–	Developed/undeveloped regions, market dynamism	Political ties have a U-shaped relationship with product innovation. This relationship is weaker in developed regions and is stronger when market dynamism is high.
Jean et al. (2018)	Resource-based theory	Product innovation	–	Competitor opportunism, technological dynamism	Political ties positively impact product innovation, and competitor opportunism positively moderates this relationship, but technological dynamism negatively moderates this relationship.
Shu, Page, Gao, and Jiang (2012)	Social network theory, organizational knowledge creation theory, knowledge-based view (KBV)	Product and process innovation	Knowledge exchange and combination	–	Political ties impact product and process innovation through knowledge exchange and combination.
Su and Yang (2018)	Opportunity-motivation-ability perspective	Exploratory innovation	–	Interaction of innovation orientation and absorptive capacity	The impact of political ties on exploratory innovation is not significant. Innovation orientation and absorptive capacity interactively and negatively moderate this relationship.
Wang, Zhang, and Shou (2019)	Dynamic capability theory, institutional theory	Market-focused innovation capability	–	Legal enforceability	Political ties negatively impact innovation capability, and legal enforceability helps relieve this negative effect.
Wu (2011)	Social network theory	Product innovation	–	–	Political ties have an inverted U-shaped impact on product innovation.
Zhang et al. (2019)	Institutional theory	Product innovation performance	Institutional support	Country (China and India)	Political ties contribute to product innovation indirectly through government support in China. This relationship is not supported in India.
Zhang et al. (2020)	Resource dependence theory, dynamic capability theory	Innovation performance	Business ties and an entrepreneurial orientation	Environmental dynamism	Political ties affect innovation performance through business ties and an entrepreneurial orientation. The indirect effects of political ties (through business ties and an entrepreneurial orientation) on innovation performance are stronger in more dynamic environments.

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